

AJAY GOWDA

Phoenix, AZ, USA

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WORK EXPERIENCE

REPUBLIC SERVICES

Phoenix, USA

Sr. Manager, Operations Technology

Sept 2023 – Present

- Evaluate and support any new fleet modernization technology.
- Led the team that built and piloted Republic's AI camera based driver safety coaching program.
- Piloted the first fully integrated prototype electric vehicle, driving its adoption through field training.
- Led a team focused on the continuous improvement and reporting of the AI refuse contamination detection system.
- Developed and designed billing process for AI refuse contamination detection system that accelerated the project by six months, generating \$40 million in revenue.
- Acted as a technical liaison between Republic's business and IT teams.

Manager, Technical Operations

Jun 2022 – Aug 2023

- Partnered with Oshkosh Corp to develop the first fully integrated electric refuse vehicle by setting customer requirements for an improved cab and safety specifications.
- Design and developed the electric vehicle viability models that dictated EV deployment locations.
- Supported the fleet electrification.
- Piloted electric vehicle pilots at different locations and compiled results to better inform fleet electrification strategy.

EVBOX

Amsterdam, NL

Systems Validation Engineer

Sep 2020 – May 2022

- Design, develop and execute automated firmware tests on current and next gen products for regression testing.
- Develop, deploy and maintain data acquisition infrastructure like databases, visualization tools and vpns on AWS.
- Design and develop one off test setups for unique test scenarios including data acquisition and generating reports.
- Design and develop test benches for correct and repeatable hardware and software validation and verification.
- Assist with pre-compliance testing for certifications such as Energy Star and UL.
- Develop and maintain EOL (End of Line) test applications.

ETERNAL SUN SPIRE

The Hague, NL

Development and Validation Engineer

Aug 2018 – Jul 2020

- Troubleshoot issues with prototype and production Photovoltaic Test Equipment in Lab and Manufacturing lines.
- Design and develop control architecture for complex machines to enable optimal performance and data acquisition.

GENERAL MOTORS

Milford, USA

Hybrid Powertrain Development Engineer

Jan 2016 – Aug 2018

- Develop efficient powertrains with optimal fuel economy and exceptional performance via work on batteries, motors, hydraulic systems, internal combustion engines, and power electronics.
- Analyze and collect data from vehicles on test tracks, in extreme locations, and state of the art high feature test cell using vehicle monitoring software and hardware.
- Validate new hybrid and electric powertrains subjectively and objectively.
- Ensure vehicle test fleet is up to date with latest hardware and software.
- Construct one-off wiring harnesses that include high voltage and CAN communication loops.

Algorithm Development Engineer – Thermal Controls

Jul 2015 – Jan 2016

- Developed thermal control system for liquid cooled High Voltage battery systems.
- Software functionality test using SIL and HIL testing techniques.

UNIVERSITY OF WASHINGTON

Seattle, USA

Research Assistant at EcoCAR Laboratory

Oct 2013 – Jun 2015

- Research Assistant at UW EcoCAR Lab.
- Worked on control logic of the supervisory controller that interlinks the two powertrains. (Electric and Diesel)

EDUCATION

UNIVERSITY OF WASHINGTON

Seattle, USA

Master of Science in ME, Control Systems - CGPA - 3.6/4.0

Sept 2013 – May 2015

- Linear Controls, Digital controls, Optimal Controls
- Master's Thesis - System Level Modeling and Simulation of a Hybrid Vehicle.

MANIPAL INSTITUTE OF TECHNOLOGY

Manipal, India

Bachelor of Science in Mechatronics Engineering - CGPA - 8.9/10

June 2009 – May 2013

- Control Theory, Analog and digital circuit design, Embedded Systems, PLC, Hydraulic and Pneumatic systems.

TECHNICAL SKILLS

Languages: Python, MATLAB/ Simulink, JAVA, C/C++, TypeScript, HTML, Structured Text (PLC), YAML, XML, SQL

Technologies: AWS, Grafana, Influxdb, Wireguard, MQTT, REST, Docker, Linux, SCADA, MODBUS, OPC, ARM, GIT, Polarion

Frameworks: FastAPI, Django, pytest, opencv

Automotive industry tools: MDA, INCA, Vehicle Spy, dSPACE, neoVI, High Performance Driver Education from GM